

STUDENT SYLLABUS

Introduction to Psychology and its Methods AI

B1, period 1, academic year 2023-2024

Course code: XB_0069
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ECTS: 6
Course load: 24.5 hours lectures (14 lectures)
2.5 hours exam
141 hours independent studying

VU University Amsterdam
Faculty of Behavioral and Movement Sciences



Information about the course

1. Introduction

The course “Introduction to psychology and its methods AI” will introduce you to the field of psychology and its methods.

2. Position in the curriculum

The course “Introduction to psychology and its methods AI” is an introductory course. There are no entry requirements.

3. Outline

This course is set up in two sections.

The first 4 weeks, we will cover the genetic and biological basis of behavior, we will discuss general human functioning, including consciousness, perception, learning, memory, thought, language, and intelligence.

The final 3 weeks, we will go through an introduction to scientific research methods: scientific reasoning, validity, measurement scales, reliability, descriptive, correlational, experimental research, within- and between-subjects designs, factorial designs, small-N and quasi-experimental designs.

4. Organization

The course runs for 7 weeks, with an exam in the 8th week. Every week, you will attend 2 lectures.

There are 2 pages for each topic on Canvas entitled ‘Questions to ask yourself!’ and ‘Something to consider’.

- **‘Questions to ask yourself!’** pages are there to provide you guidance on what questions lecture you should be capable of describing and understanding.
- **‘Something to consider’** section is optional. (contains elaborations on a specific (related) issue and is meant to provoke some further reflections on the corresponding topic.)

The present syllabus provides you with an overview of these topics. For each topic you’ll find:

- **Description**
 - a very brief description of the topic
- **Literature**
 - outlines the corresponding required readings

Correspondence with the coordinator

Most information about the course will be available on Canvas. For remaining questions, it is possible to contact the lecturer personally (d.g.oezkan@vu.nl).

5. Literature

Gazzaniga, M.S. (2018). Psychological science (7th edition). Norton.

Morling, B. (2021). Research methods in psychology (4th edition). Norton.

6. Examination

- **Final exam:** Both the specified pages of the books in the schedule below, and lecture slides will be included in the exam. The exam consists of 56 multiple choice questions. These are difficult questions that often go into detail. There are two opportunities to take the exam (see the VU course schedule for definitive locations):
 - Final exam: 21st October 15:30
 - Resit exam: 7th January 12:15
- **Grading**

All multiple choice exam grades are chance corrected.
 Passing grade: 5.5*

*Note that the raw exam scores are recalculated through chance correction. (The cut-off score of 35 out of 56 questions equal the passing grade of 5.5, instead of 31.)

7. Learning objectives

After successfully completing the course, the student is able to:

1. Understand main theories and concepts in psychology, including perception, cognition, behavior, and learning.
2. Identify and describe various research methods used in psychology, such as experiments, correlational studies, surveys, and observational studies.
3. Understand basic principles of psychological measurement and assessment, including reliability, validity, and different types of scales (nominal, ordinal, interval, ratio).
4. Understand how empirical research allows us to deduce things about the mind
5. Understand the differences between common misconceptions and psychological research

8. Schedule

Day	Date	Time	Lecture No	Topic	Location(s)	Reading
Mon	2/Sep	11.00 - 12.45	L1	Genes and Evolution	MF-FG1	Gazzaniga, Ch1: 3-16; Ch3: 102-116; 2 links provided on Canvas (8 pages)
Thu	5/Sep	13.30 - 15.15	L2	The brain and the nervous system	MF-FG1	Gazzaniga, Ch3: 67-101
Mon	9/Sep	11.00 - 12.45	L3	Consciousness	MF-FG1	Gazzaniga, Ch4: 119-158
Thu	12/Sep	13.30 - 15.15	L4	Sensation and Perception (Vision)	MF-FG1	Gazzaniga, Ch5: 159-186
Mon	16/Sep	11.00 - 12.45	L5	Learning	MF-FG1	Gazzaniga, Ch6: 203-241
Thu	19/Sep	13.30 - 15.15	L6	Memory	MF-FG1	Gazzaniga, Ch7: 243-281
Mon	23/Sep	11.00 - 12.45	L7	Thinking and Language	MF-FG1	Gazzaniga, Ch8: 283-304; 314 -327
Thu	26/Sep	13.30 - 15.15	L8	Intelligence	MF-FG1	Gazzaniga, Ch8: 304-314; 2 videos provided on Canvas
Mon	30/Sep	11.00 - 12.45	L9	Scientific reasoning	MF-FG1	Morling, Ch. 1 and 2
Thu	3/Oct	13.30 - 15.15	L10	Validity	MF-FG1	Morling, Ch. 3 and 11 (p. 322-341)
Mon	7/Oct	11.00 - 12.45	L11	Measurement, reliability, and descriptive research	MF-FG1	Morling, Ch. 5 (p. 116-136), Ch. 6, and 7 (p. 178-189)
Thu	10/Oct	13.30 - 15.15	L12	Correlations and experiments	MF-FG1	Morling, Ch. 8, 9 (p. 240-248), 10, and 11 (p. 342-357)
Mon	14/Oct	11.00 - 12.45	L13	Single-factor and factorial designs	MF-FG1	Morling, Ch. 10 and 12
Thu	17/Oct	13.30 - 15.15	L14	Small-N and quasi-experimental designs	MF-FG1	Morling, Ch. 13
Mon	21/Oct	15:30 - 17:45	Final Exam			
Tue	07/Jan	12:15 - 14:30	Resit Exam			

Topic 1: Genes and evolution

Description

This lecture will shortly introduce what psychology is. In addition, the genetic and evolutionary basis of behavior will be discussed.

Topic 2: The brain and the nervous system

Description

This lecture will focus on the biological basis of behavior: the nervous system. The knowledge you'll obtain in Week 1 is important, since we will often refer back to it during the rest of the course.

Topic 3: Consciousness

Description

We will discuss the meaning of consciousness and the different kinds of consciousness after brain damage. We will also talk about variations in consciousness, e.g. those induced by changes in attention, sleep, hypnoses, or drugs.

Topic 4: Sensation and perception: vision

Description

We will look at both early (sensory) processes and later (perceptual) processes. Eventually, it will become clear that perception is a complex process in which the brain does not just passively receive information from the senses, but also actively influences this information (interpretation).

Topic 5: Learning

Description

We will mainly discuss the characteristics and logic of classical and operant conditioning. We will also briefly look at non-associative learning and observational learning.

Topic 6: Memory

Description

We will look at the different sub-processes of memory, including encoding, storage, and retrieval of knowledge. We will also pay attention to memory errors and the different kinds of memory that can be identified.

Topic 7: Thinking and language

Description

The first part of the lecture about thinking and language will be dedicated to three aspects of thought: mental representations, decision making, and problem solving. The second part will focus on language.

Topic 8: Intelligence

Description

We will not only discuss what intelligence is and how we can measure intelligence, but also to what extent intelligence is determined by genetic and / or environmental factors.

Topic 9: Scientific reasoning

Description

This lecture provides an introduction to the logic behind science, scientific reasoning, and how scientific statements are different from claims based on personal experience, intuition, and authority.

Topic 10: Validity

Description

This lecture focuses on validity, i.e., construct validity, external validity, internal validity, and statistical validity.

Topic 11: Measurement, reliability, and descriptive research

Description

This lecture will introduce you to the different measurement scales and the concept of reliability. Following the focus will be on the two major forms of descriptive research: surveys and observational research.

Topic 12: Correlations and experiments

Description

This lecture is concerned with correlational research and the basics of experimental research.

Topic 13: Single-factor and factorial designs

Description

This lecture will be primarily concerned with single-factor and multiple-factor (factorial) designs.

Topic 14: Small-N and quasi-experimental designs

Description

In the last lecture, we will look at designs with small numbers of participants (small-N designs) and designs used in experiments in which researchers do not have full experimental control (quasi-experimental designs).